Chapter 1
Categorical Data Analysis

Categorical data analysis refers to statistical analysis that studies variables and data whose measurement scale is primarily categorical or discrete. In general, the main variable, which is a categorical variable, is commonly called the response variable, and its observed values are counts, referred to as categorical data, which may be independent or dependent, as in repeated measures, under distributions belonging to the exponential family or the non-exponential family, as well as involving fixed effects and random effects, with explanatory variables or other related variables being categorical, continuous, or both.

In addition to the main variable of interest, categorical data analysis also aims to study the structure of relationships among variables, particularly the explanatory variables that have influence (effects) or impact on the various levels of the response variable. The sources of data may come from experiments or from sample surveys that serve as good representatives of the population, and the results can be analyzed and inferred to the population. Examples include the analysis of count data that appear in the form of contingency tables, the determination of the magnitude of association between categorical variables, the testing of relationships between variables, and the construction of statistical models, both simple and complex, for categorical response variables and various explanatory variables. Statistical models, which include models with a probabilistic component, can be used to evaluate outcomes such as goodness of fit, the factors that affect the response variable, the prediction of group probabilities, as well as other statistics.

In general, categorical variables may classify groups according to natural characteristics, such as the variable “gender” or the variable “rank.” However, some categorical variables may be transformed from continuous data into grouped data that carry more meaningful interpretation according to international standards or expert criteria. For example, a categorical variable related to students’ academic performance may be organized or transformed according to students’ cumulative grades throughout the entire program into four categories or levels, such as: honors, average, and others, in order to directly study educational categorical achievement. A categorical variable related to disease status may need to be diagnosed as belonging to the severe, moderate, or normal category or level, based on measurements or examination results that are continuous variables, according to the criteria of medical experts. This is done in order to make decisions on selecting appropriate treatments for a particular disease status in a timely manner and to prevent adverse effects on the patient.

If the variables and data in which the researcher is interested are not originally categorical in nature, it is generally preferred to group such data so that they correspond to categorical variables first. This allows the use of appropriate categorical data analysis techniques and enables problems to be addressed accurately. Such grouping or transformation of data does not represent a reduction of information, but rather a way to enable the use of categorical data analysis techniques, which are statistical methodologies that have been progressively developed and theoretically supported.

<u>Terms of Reference</u>

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1.1 Types of Data

Data refers to a set of observed values measured from a certain characteristic. They may be numbers, characters, text, information, images, or symbols with specific meanings, which are then processed, interpreted, or used in procedures for obtaining additional information.

The classification of data, for convenience in selecting appropriate methods of data analysis, may categorize data by their characteristics as follows:

1.1.1 Data classified according to <u>the properties of the data</u> are of two types:

1. **Quantitative data** are data obtained from observation or measurement expressed as numerical values. In general, they are continuous data, such as weight, height, blood sugar level, crop yield weight, and plant harvesting age, and so on.
2. **Categorical data:** If the observation or measurement of each unit is classified into one category among several categories under study, the data in each category of this dataset are called categorical data, or in some places referred to as qualitative data. In general, they are count data, which are discrete data, such as counts of data divided by different groups, for example, gender groups, color groups, rank levels, levels of work performance, levels of occurrence of a certain disease, levels of attitude, plant varieties, seed characteristics, and so forth.

1.1.2 Data classified according to the nature of **<u>data collection</u>** are of four types as follows:

1. **Survey data** are data obtained from surveying information from a population or from a sample. If the data come from the entire population, it means that the information is surveyed from all units of the population. If the data come from a sample, it means that the information is surveyed from some units of the population. This survey is a process of collecting data that depends on the method of surveying and the method of data collection, which will be discussed later. For example, surveying the status of various industrial factories—if the survey is conducted at every factory, the data obtained are survey data from surveying every unit in the population and are called census data. Similarly, if data are collected from only some factories through a sampling method under a specific survey method or a specific sampling design, or under a specific sample selection procedure, they are called sample survey data, which may

be surveyed with or without the use of probability in selecting the sample.

2. **Experimental data** are data that must be obtained through some form of experimentation, with consideration of the representativeness or randomness (randomization). For example, in engineering, one may measure the length of steel bars tested in a laboratory, or examine the rate of occurrence of flaws or defects in an experimental process or in the production of a certain type of product. In science, the experiment may involve agricultural science or experiments concerning plants, animals, or chemicals. In medicine, it may involve experimental studies in epidemiology and methods of treating disease. In the social sciences, experiments may study various human behaviors, and other areas. The data obtained from experiments may come from observation or from measuring the outcomes of the experiment and are then examined and analyzed further.

1.1.3 Data classified according to **<u>the sources of data</u>** are of two types:

1. **Primary data** are data obtained directly from the original source of the information. Users of this type of data must collect the data themselves. Examples include data that researchers obtain directly from census units or survey units, or outcome data obtained from experiments conducted by the researchers, such as testing a new drug and an old drug on laboratory rats. These are not data previously collected by others or taken from other sources.
2. **Secondary data** are data that are not observed or measured directly from population units or samples, but are obtained from others or from documents previously produced. Examples include data obtained from registration records, from past research reports, from books, from websites, and so forth.

1.1.4 Data classified according to the **<u>types of measurement scales</u>** are of four types:

General data classified according to the measurement scale are as follows:

1. **The nominal scale, or nominal data,** is the type of data with the lowest level among the four types. It is data that indicate classification or grouping, showing to which group each unit belongs and how many units are in each group. Examples include the number of units classified by gender, administrative region, or geography. Nominal data may also be general data that do not represent counts but only indicate differences, such as postal codes, bus route numbers, and telephone numbers.
2. **The ordinal scale, or ordinal data**, is data with a higher level than the first type. In addition to being able to classify or distinguish groups, it can also be ordered from low to high or conversely from high to low. For example, the academic ranks of university instructors may be ordered as professor, associate professor, assistant professor, and lecturer. Military ranks in the army may be ordered as general, colonel, and captain. Educational levels may be ordered as doctoral, master's, and bachelor's degrees. Each group may contain different numbers of people or units of interest, which are the data we want to know. Additionally, a survey of preferences for five items may ask respondents to rank them from 1 to 5, where smaller numbers indicate lower preference than larger numbers. However, the numerical values, for instance, 2 and 4, do not mean that preference level 2 is half as strong as preference level 4, or that the rank of general is higher than that of colonel by the same amount that a colonel's rank is higher than that of a captain. Similarly, a patient's condition may be classified as very severe, severe, moderate, mild, or normal, or the evaluation outcomes of an organization may be classified as excellent, good, normal, needs improvement, or poor in order to use these results for further action.
3. **The interval scale** is data with a higher level than the previous types. In addition to being able to classify and order data, interval data allow us to indicate the magnitude of the difference between the ordered values, that is, how far apart the intervals are. Examples include temperature data. For instance: at noon today, the temperature in Nakhon Pathom is measured at 30°C, while in Kanchanaburi it is measured at 40°C. This type of data does not begin measurement from zero; that is, it does not have a natural zero.

A temperature value of zero does not mean the absence of temperature but rather that the temperature is truly at 0°C, and temperature can take negative values.

Therefore, from the above example, it can only be stated that the temperature level in Kanchanaburi is higher or greater than the temperature level in Nakhon Pathom by 10˚C. However, the temperature in Kanchanaburi is **not** $\frac{40}{30}=1.33$ times that of the temperature in Nakhon Pathom, because such temperatures cannot be compared as ratios.

4. **The ratio scale, or ratio data,** is data with the highest level among the four types. It allows us to state differences, order values, measure interval distances, and make ratio comparisons. Ratio data have a natural zero and are commonly scientific data such as weight, length, width, depth, and various quantities.

The above types of measurement scales may be combined into only **two categories** as follows:

1. **Qualitative data** consists of the two types of data mentioned above, namely nominal data and ordinal data. These are collectively referred to as qualitative data or categorical data. Such data is non-continuous and is expressed as counts classified according to qualitative categories.
2. **Quantitative data** consists of the latter two types of data, namely interval-scale data and ratio-scale data, and these are collectively referred to as quantitative data. In general, such data is continuous, representing quantities measured directly from sampling units or experimental units.

Therefore, once the objectives, the variables, the characteristics of the various variables, and the data of interest are clearly identified, the next step is to consider the method of data collection. This involves first determining whether the data will be collected through a survey, through an experiment, or whether a survey should be conducted first before carrying out an experiment on the sampling units. Each approach has important distinguishing criteria, such as:

- **If the data are obtained from a survey,** especially when the aim is to obtain information from a representative sample of the entire population, it is necessary to employ appropriate sampling techniques, referred to as **sampling designs**, to assist in planning the data collection procedure.
- **If the data are obtained from an experiment**, statistical techniques known as **experimental** designs should be used instead of sampling designs.
- **If the data are of a mixed type, involving both sampling surveys and experimental procedures,** techniques relevant to each stage of data collection may be applied jointly. For example, an appropriate sampling design may be chosen, along with a suitable method of experimental planning, to be used together in the data collection process of interest.
- **If the data come from administrative records or other preexisting sources,** whether they are census data, previous survey data, or experimental data collected by others, the method of analysis will depend on the design originally used. However, if the origin of the data is unknown, or if it is historical data with unclear provenance, the analysis should be based on the simplest possible method.

Moreover, regardless of whether data are collected through surveys, experiments, or other sources, which each involve different sets of criteria, every method of data collection requires certain principles that rely on measurement methods (measures), interviews, counting, or observation of data. Examples include the use of medical examination instruments, achievement tests, various measuring devices for weighing or quantifying, observational procedures, and interview forms, among others. The researcher may select one such method, which may be summarized as follows:

1. **The use of instruments:** For example, measuring blood pressure requires a medical device for blood pressure measurement; measuring the quantity of a chemical substance requires a chemical measurement instrument; and likewise, measuring weight, height, temperature, speed, or academic achievement through examinations.

The use of such measuring instruments is generally applied primarily in the collection of experimental data. The form used for this purpose is a data recording form.

2. **Observation or direct counting from population units or sample units:** For example, observing the number of people entering and leaving a department store during a given period; counting the number of individuals using an automated teller machine of a particular bank during a certain time interval; or observing a particular type of work behavior. This method of observation is generally used for collecting data from surveys, and the form used is a data-recording form.
3. **Interviews:** This is a widely used method of data collection in the social sciences, business, public health, psychology, education, and other fields. The form used for interviews is called a questionnaire or a schedule. Interviews are conducted with population units or various sample units, and interviewers must be systematically trained beforehand. Examples include interviewing individuals regarding their opinions on the election of members of the House of Representatives, interviewing individuals about their preference for a particular product, and conducting interviews not only in person but also by telephone, or by sending questionnaires through the postal service as an alternative method. In general, interviews are used for data collection in surveys.
4. **From records and other documents:** This is a method of data collection that does not obtain data directly from population units or sample units, but rather from documents already prepared. Thus, the data collected are secondary data used anew. Examples include transcribing information from research reports or various other reports. Data obtained by this method must be assessed for accuracy and timeliness to ensure the usefulness of the information. For instance, registration records of migration, marriage records, or other types of registries must be examined to determine whether they are outdated and to what extent they are reliable.
5. **The use of the data collector's judgment:** This method is similar to observation or counting, but it requires the observer's judgment in determining what the obtained data represent, what type they are, or

how much they amount to. The data may consist of approximate numerical values based on the observer's experience. Examples include observing the clothing styles of men and women in a particular society, assessing language pronunciation by listening to accents, such as evaluating English speaking practice, and observing the amount or characteristics of daily rainfall. In general, these methods require that the data collector be a person with expertise in the subject of observation. Training someone else to carry out such observations in their place is a process that typically requires considerable time.

Summary of the principles of data collection: Data collected from sample surveys and from experiments customarily employ the principles of sampling theory (sampling designs) and experimental design, respectively, to assist in planning the data collection process under an appropriate design. Examples include a multistage sampling design or a completely randomized design, among others. In cases where both sample surveys and experiments are used, the designs of both types may be applied in sequence.

Once the survey or experimental plan has been properly established, the collection of data from sample units or experimental units may involve the use of measurement instruments, interviews, observation methods, and so on. Before processing and analyzing the data obtained, the types of variables and data, as well as the various properties of the data, should be examined in order to help us select analytical methods that are appropriate for the data analysis techniques to be applied.

1.2 Categorical Variables and Categorical Data

Categorical variables are defined as follows:

> **Definition 1.1. Categorical variables** are variables whose possible values fall into one of several groups that differ in qualitative characteristics, with at least two such groups. The scale of measurement consists of a set of non-numeric categories. In general, categorical variables are of two types: nominal variables, such as gender, product brand, and political party; and ordinal variables, such as levels of attitude, levels of disease severity, and various levels of effectiveness. In addition, continuous variables may be transformed into categorical variables and categorical data according to universal criteria or expert judgment.

Agresti (2002) expanded the meaning of categorical variables to include the following types of variables:

(1) Nominal variables
(2) Ordinal variables
(3) Discrete interval having relatively few values
(4) Continuous variables grouped into a small number of categories

For example, the smoking variable classified as never smoked, formerly smoked, and current smokers,
the income variable classified as low, medium, and high,
the product variable classified as brand A, brand B, and brand C,
the securities performance variable classified as continuing operation, temporarily suspended, and closed down

In summary, the measurement scale of categorical variables corresponds directly to discrete variables. However, if the variable is continuous, it can still be categorized (categorization) by converting a continuous variable into a discrete one (discretization of the continuous variable), and the counts in each group can be observed. For example, the high blood pressure variable, which is measured as a continuous variable, can be transformed by grouping according to expert criteria into categories such as very high blood pressure, moderate, and not high. Then, by counting the frequency of data classified according to those categories or groups, one obtains a categorical variable and categorical data. At present, such data are widely used particularly in biomedical science and social science, and, in fact, not only in these two fields. Other disciplines also commonly use grouped data and techniques for analyzing categorical data, applying them appropriately, such as econometrics, economics, psychology, education, quality control, public health, pedology, engineering, computer science, agriculture, environmental science, history, and geography.

The observation of categorical data, or the measurement of categorical data, is defined as follows:

> **Definition 1.2 Categorical Data** refers to data that consist of the counts of observations that fall into one group or another, either in nominal form or in ordinal form, including grouped data transformed from interval-scale or ratio-scale data according to universal criteria or expert standards.

The distinction between categorical variables that are response variables or dependent variables and categorical variables that are explanatory variables or independent variables should be considered as follows:

1. **Data concerning the experimental units or sample units** that arise as outcomes during an experiment or survey. If one is interested in applying a statistical model to analyze the data, such as the logit model, this type of data corresponds to the response variable or dependent variable of the model.
2. **Data concerning the groups** of experimental units or concerning the conditions of interest that are assigned to the experimental units or sample units. If a statistical model is used to analyze such data as described above, this type of data corresponds to the explanatory variable or independent variable of the model.

The general characteristics of categorical variables and categorical data have forms that allow the classification of the factor combinations of the response variable and the explanatory variable, as shown in Table 1.1.

Table 1.1 General characteristics of categorical variables and categorical data

Categories of the explanatory variable	Categories of the response variable				Total
	1	2	…	c	
1	n_{11}	n_{12}	…	n_{1c}	$n_{1\cdot}$
2	n_{21}	n_{22}	…	n_{2c}	$n_{2\cdot}$
3	n_{31}	n_{32}	…	n_{3c}	$n_{3\cdot}$
⋮	⋮	⋮	⋮	⋮	⋮
r	n_{r1}	n_{r2}	…	n_{rc}	$n_{r\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	…	$n_{\cdot c}$	$n_{\cdot\cdot}$

n_{ij} represent observations or counts, where $i = 1, \ldots, r,$ and $j = 1, \ldots, c$

$n_{\cdot\cdot}$ represent the total number of samples

Examples of categorical variables and categorical data are shown in Table 1.2.

Table 1.2 Lung cancer classified by daily smoking history

Smoking history	Disease status (Y)		
(X)	Lung cancer	others	Total
none	5	38	43
low	8	47	55
medium	17	48	65
high	7	9	16
Total	37	142	179

It should be noted that the data in a contingency table, besides being counts or frequencies, may also be presented in the form of probabilities or proportions, such as the data in Table 1.3, presented by Ku & Kullback (1974). The data are from physical examinations of 1,329 American males, and it was found that the proportions of patients with heart disease classified by blood pressure (BP) levels and cholesterol levels are as follows:

Table 1.3 Proportion of heart disease patients among examinees classified by blood pressure level and serum

Cholesterol (CHOL) level (mg/100 cc)	Blood pressure (BP) level			
	127	127 − 146	147 − 166	167
< 200	2/119	3/124	3/50	4/26
200 − 219	3/88	2/100	0/43	3/23
219 − 259	8/127	11/220	6/74	6/49
> 259	7/74	12/111	11/57	11/44

The variables used in the analysis of categorical data can be summarized from Section 1.2 by classifying the types of categorical variables according to the purpose of categorical data analysis as follows:

1. Response and Explanatory Variables

In the analysis of categorical data, it is customary to consider the types of variables first, focusing on the main variables of interest. If one intends to use a statistical model to assist with the data analysis, the main issue or variable under investigation is treated as the response or dependent variable which, according to theory, is a random variable, while the other related variables serve as explanatory or independent variables, or non-random variables. This is similar to most statistical data analyses that distinguish between response variables and explanatory variables, such as regression models that explain changes in the mean of the dependent variable based on the values of the independent variables. In categorical data analysis, the response variable is always a categorical variable, while the explanatory variables may be of any type. Appropriate and reliable models and statistics must be selected accordingly.

For example, when applying logistic models to study events related to survival after heart surgery among heart disease patients, let the response variable (Y) represent the survival or death of the patient, and study the explanatory variables (X_i) that may help explain the survival outcome after heart surgery. These include three variables:

change in age (X_1), cholesterol level (X_2), and the type of heart disease of the patient (X_3).

In summary, the response variable in the analysis of categorical data, as referred to in this book, means a categorical response variable. As for the explanatory variables, in most methods of categorical data analysis, they are categorical, and the data can be analyzed in the form of contingency tables. However, the data of the explanatory variables may be continuous, but appropriate statistics must be selected. In cases where an explanatory variable is continuous, it may be transformed into a categorical variable based on expert criteria. Therefore, the explanatory variables used in models for categorical data analysis may be of both types, and the appropriate method of analysis and statistical procedures should subsequently be selected.

2. Categorical variables according to the categorical measurement scale

The measurement scales for categorical data analysis generally consider the first two or three types, namely "nominal," "ordinal," and "interval," as mentioned in Section 1.1.4, as follows:

Nominal variables: Categorical variables whose categories are not naturally ordered or cannot be arranged according to any meaningful hierarchy. Examples include the variable of religion (Buddhist, Christian, Muslim, others) and the variable type of residence (single house, townhouse, condominium, others).

Ordinal variables: Categorical variables whose categories or levels can be arranged in a natural order, but the distances between the ordered levels are unknown. Examples include the variable of engine size (small, medium, large); the variable military rank (below third class, second class, first class, colonel, general); and variables of attitudes or opinions such as abortion (strongly disagree, disagree, no opinion, agree, strongly agree).

Discrete interval variables: Categorical variables whose categories or levels can be measured by the distances between the ordered levels, **which are discrete and have only a limited number**

of values (discrete intervals having relatively few values). Examples include the variable blood pressure and others, as shown in Table 1.4.

Therefore, for categorical data analysis, the response variable is often a nominal or ordinal variable and can also be applied together with Interval-type variables, such as the variable concerning the number of times a person has been married, which may be grouped (never married, married once, married twice, married more than twice), or educational variables such as years of schooling (less than 10 years, 10-12 years, more than 12 years), and variables representing different levels of opinion, and so forth. These variables resemble temperature variables in that they do not have a natural zero. For example, an opinion score of zero does not mean "no opinion," but rather may indicate that the respondent has some opinion but prefers to abstain or not express it. Likewise, a length of study equal to zero does not imply being totally uneducated; similarly, a television with a service life of zero years does not mean it has no lifespan, but may mean that the device malfunctioned so quickly after beginning use that it had to be discarded (for instance, a service life of less than one year assigned as 0, and a service life of 1 to less than 2 years assigned as 1, and so on). Being married for zero years does not mean not being married; it may mean just recently married, or other interpretations depending on the intended meaning, or other.

As mentioned above, the measurement of categorical variables, which are discrete in nature, arises from various limitations of the measuring instruments. Therefore, in general statistical data analysis, some interval variables that have sufficiently large amounts of data or sufficiently many numerical values are often treated as continuous variables, such as examination scores, and may be analyzed according to the methods for continuous variables. However, if the data clearly exhibit the characteristics of a discrete interval, the data may instead be analyzed according to the methods for categorical variables.

In summary, categorical variables may have the following types of data:

(1) Nominal, for example, the variable gender, with 2 groups: 1. male 2. female, presenting how many people are in each group

(2) Ordinal, for example, the variable academic year level, which is 1. first year, 2. second year, 3. third year, 4. fourth year, presenting how many people are in each group

(3) Discrete Interval, which has only a few numerical ranges, such as the variable number of children in a household:

1. fewer than 5 persons 2. fewer than 10 persons 3. 10 persons or more, presenting how many people are in each group

For the case of Continuous Interval, transformation may be applied by grouping the numerical values into a few categories, such as monthly income groups or salary levels of employees in a company, where the data of interest are observations, counts, or frequencies, as in Table 1.4.

Table 1.4 Examples of categorical variables of the Discrete Interval type and the Continuous Interval type

Discrete Interval (Years of schooling: years)	Data (counts)	Continuous Interval (Salary of a company: baht)	Data (counts)
1. ≤ 6 years	40	1. $5,000-18,999$ baht	35
2. $7-12$ years	50	2. $19,000-28,999$ baht	55
3. $13-15$ years	30	3. $29,000-38,999$ baht	20
4. >15 years	20	4. $39,000-48,999$ baht	15
		5. $49,000-59,999$ baht	10
		6. $>60,000$ baht	5

Nominal variables are qualitative variables, while Interval variables are quantitative variables. For Ordinal variables, in practice, they are commonly treated as qualitative variables. However, for convenience and to allow multiple analytical options, some researchers assign numerical values to the groups of qualitative data at different levels in order to give them quantitative characteristics. This method must be used with caution and should be carried out under the supervision of experts and researchers to ensure that the numerical transformation does not distort the meaning or cause other effects, especially in the interpretation of the data.

The symbols of the primary variables of interest, such as the response variable, are represented by or (the expected frequency) or other notations depending on the nature of a particular model. As for the characteristics of the explanatory variables, whether they are quantitative or qualitative, they are represented by ***X***. If there are multiple variables, categorical variables will take the form of factors, especially in the chapter on loglinear models (Chapter 8), which concerns data tables composed of several categorical variables without regard to whether they must be response variables or explanatory variables. Instead, the expected values of each category of the various variables, (m_{ijk}) serve as the response variable of the model involving three variables, such as X, Y, and Z. The analysis may aim to predict expected values, examine associations, or investigate joint effects among categorical variables, as well as the influence of variables in terms of various odds, and so on.

Example 1.1 Characteristics of single-variable categorical data, such as the number of people or observations in each group classified according to a categorical variable, are as follows: the birthplace of first-year students in the Faculty of Science for the academic year 2009 can be categorized according to different provinces, as assumed in Table 1.5

Table 1.5 Number of new students classified by birthplace

Birthplace of new students	Number of new students, Faculty of Science, 2009
Bangkok	120
Nakhon Pathom	100
Ratchaburi	80
Suphan Buri	90
Others	300
Total	690

If each observational unit can be assigned to sets of groups in multiple ways, in other words, if each unit can be classified in, for example, two ways (Two-way classification), this is called bivariate categorical data. An example of such data is shown in Table 1.7.

Example 1.2 From the following categorical data, we wish to classify the characteristics of bivariate categorical data by assuming that we are interested in the variables color and shape of sports equipment, with 6 types, one item each, as shown in Tables 1.6–1.7.

Table 1.6 Categorical data on the color and shape of sports equipment classified by type of equipment

Type of equipment	Color variable	Shape variable	Number
American football	brown	oval	1
baseball	white	round	1
golf ball	white	round	1
basketball	brown	round	1
badminton	white	pointed cone	1
hockey	black	round cone	1

From Table 1.6, the data on color and shape of the equipment can be grouped into categories of color and categories of shape, forming bivariate categorical data as shown in Table 1.7.

Table 1.7 Bivariate categorical data classified by the variables color and shape of sports equipment

Shape of sports equipment:	**Color of sports equipment**		
	black	brown	white
oval	0	1	0
round	0	1	2
pointed cone	0	0	1
round cone	1	0	0

From Table 1.7, each category appears only once. The total data for each type of equipment sums to 6 items, which is called a two-way

table or two-way contingency table. If the data are classified by more than two variables, it becomes a multidimensional contingency table, which is not discussed in this chapter.
The characteristics of categorical variables and examples of categorical variables are summarized as follows:

Table 1.8 Summary of the characteristics of categorical variables of the discrete type and the continuous type

Characteristics of variables	Examples of categorical variables:
Categorical with 2 groups (Binary categorical)	male/female state, enterprise/private, fair-skinned/yellow-skinned, military/police, present/absent, recovered/not recovered, successful/unsuccessful, took medicine/did not take medicine
Categorical with >2 groups (Nominal scale)	blood type, cause of death, types of heart disease, color groups, religion, political party, name of university
Categorical with >2 groups (Ordinal scale)	severity level of a disease, work performance level, degree level, military–police ranks, academic year level
Discrete	number of children in a family, number of accidents at an intersection, number of dairy cows that die per year, number of automobiles
Continuous	Height, weight, birth rate, time/unit, income, children's IQ, annual rainfall

In summary, continuous variables are generally measured using quantitative data, while discrete variables are generally measured using qualitative data. For categorical variables of the nominal and ordinal types, they are generally classified as qualitative variables, which may be response variables or random variables, or may be explanatory variables used to study whether they influence the response variable. Categorical explanatory variables may be called factors, and each **factor** contains various categories, which may be called **levels**. For example, gender has 2 levels, as shown in Table 1.9, which includes a nominal gender variable and an ordinal attitude variable, with categorical frequency data shown in Table 1.9.

Table 1.9 Two-way contingency table classified by gender and attitude

Gender	Attitude toward abortion*			Total
	1	2	3	
Male	11	14	6	31
Female	32	28	29	89
Total	43	42	35	120

* 1 represents strongly agree, 2 represents moderately agree, 3 represents disagree (Lawal, 2003)

1.3 Population and Sample

Population refers to a set of data that includes all units exhibiting certain information or natural characteristics (variables) that we are interested in studying. In other words, a population is the set of all units or all data that we are interested in within the specified scope of study. The natural characteristics of the population units, such as the employment status of Thai citizens—where the population units exhibit all possible employment statuses completely—are called the population.

A population does not necessarily refer to people. In statistics, a population is a set of units of measurement (a set of measurements) that we are interested in studying, surveying, or experimenting on. It may consist of people, animals, objects, plants, factories, areas, or other measurement units that exhibit the characteristics or information we wish to study. There are two types of populations: a **finite population**, which is one in which the total number of units within the specified boundary can be known; and the other type is **an infinite or innumerably large population**, in which the total number of units cannot be completely counted, such as the number of birds in a forest or the number of fish in a body of water, and so on.

Definition 1.1 A population is the set of units that we are interested in studying. A parameter refers to a numerical value of interest that directly describes a characteristic of the population, such as a proportion (P), a mean (μ), a variance (σ^2), or other.

Example 1.3 From the U.S. census conducted in 1999 (Watkins et al., 2004), the data of interest concern the number of children per family, with the proportions of families as follows:

Number of children per family	Proportion
0	0.505
1	0.203
2	0.191
3	0.073
4 or more	0.028

The above proportions are estimates or parameters—why, and how should the data be analyzed?

Solution All of the above proportions are parameters of proportions. They are used to describe the characteristics of American families, in terms of the number of children per family, at the population level for the entire country.

✈

It can be seen that 50.5 percent of American families have no children, and 2.8 percent have ≥ 4 children. These are parameter values, not estimates from a sample, and descriptive statistics are used rather than inferential statistics, if we are unable to study the data at the full population level.

It is common to study data and information from a sample instead. A sample is a subset or a set of data selected from a population, being a part of the population. For example, a sample of 2,000 Thai citizens showing various employment statuses selected from the Thai population according to specific criteria; the number of heart disease patients selected as a sample from the population of patients in a particular hospital; or the number of experimental units used in a laboratory, where the sample units or experimental units are obtained through a probability-based method, meaning that the sample possesses the property of being representative of the population (random or randomization). Statistical inference techniques can therefore be used to assist in analyzing the data in order to study the characteristics of the population from which the representative sample was drawn appropriately.

Moreover, if the selected sample is not representative of the entire population, for example, when sampling is done by purposive sampling in order to study certain information, or when only a certain number of observations are selected to examine specific characteristics (quota sampling), the general data analysis techniques should rely on descriptive statistics to summarize the characteristics of the sample directly. Inferential statistical techniques should not be used to draw conclusions about the population from such a sample, because the sample may not be random or may not be a good representative, as certain characteristics of the population may have no chance of being included in the sample at all.

From the meanings of population and sample, it is found that if the sample possesses the property of being a good representative, then inference to the population can be made using inferential statistical methods, such as finding confidence intervals, constructing models, testing hypotheses about parameters of interest, and so forth, and summarizing the informational results in that manner under appropriate statistical techniques, from the sample data to the population.

Definition 1.2 A sample is a subset or a set of data selected from the population of interest. **A statistic** is a formula or a numerical measure that can be computed from the sample data, such as the sample proportion $(\hat{p})$, the sample mean $(\bar{X})$, the sample variance (s^2), which are statistics used to estimate the parameter values of the population proportion (p), the population mean (μ), and the population variance (σ^2) respectively.

If we pose the question from Example 1.3 of how a statistic differs from a parameter, and whether the data should be analyzed using descriptive

statistics or inferential statistics, the answer is as follows: if the data in Example 1.3 are not population data but are data surveyed from only a portion of Americans that serves as a representative sample of the population, then the proportions obtained from the sample data will be statistics, no longer parameters. Therefore, the analysis of the statistics should use inferential statistics, such as hypothesis testing, finding confidence intervals, and so on.
Before analyzing the data, the type of data as well as the source of the data should be examined to assist in selecting an appropriate and correct method of data analysis.

1.4 Categorical Data Analysis and History

Categorical data analysis is a statistical analysis technique that has been used for a long time, but in the past it was rather limited to the analysis of two-dimensional contingency tables due to the limitations of calculators; computations were usually restricted to the Chi-square statistic. Later, over the past 25 years, categorical data analysis and its applications have become more widely recognized, with many statisticians developing categorical data analysis techniques and extending continuous-data analysis techniques into categorical-data analysis. They have also expanded the limitations of variables, statistics, methods of parameter estimation for models, and two-dimensional contingency tables into generalized cases (generalization) used for multidimensional contingency tables, as well as developed more complex statistical models, such as generalized linear models, by extending the distribution of the response variable to include non-normal distributions, especially various types of discrete or categorical distributions within the exponential family distribution. Thus, GLMs encompass linear models (LMs), such as regression models and factorial (ANOVA) models. The generalized linear models that have been extended consist of three components: the random component of the model, that is, the distribution may take several forms; the systematic component, that is, the linear predictor of the model; and the link function component, that is, a function of the mean of the model. Examples of GLMs are as follows:

Logistic models or models for binary response

Logistic growth curve models such as Probit models
Log-linear models such as Hierarchical log-linear, Partial association log-linear, Direct, Generalized independence models
Logit and multinomial response models such as Multicategory response models, models for rates
Models in ordinary contingency tables such as the R+C association model
Logit models for cumulative distribution such as proportional odds models, generalized cumulative logit models
Models for matched pairs such as quasi-symmetry models
Generalized linear mixed models for categorical response such as multivariate random effects models for binary data, beta-binomial models

From the GLMs that have been extended from the LMs and LMMs by allowing the distribution of the response variable to take various forms under different link functions, further research and development of GLMs have introduced even more complex forms of relationships, so that they may be applied to real data in certain fields, particularly concerning fixed effects and random effects, which are called random-effect models.

The aspects related to the construction of statistical models may be summarized as follows:

1. Simple linear regression model by Sir Francis Galton in the early 19th century
2. Multiple linear regression model by Legendre and Gauss in the early 19th century
3. Analysis of variance (ANOVA model) by Fisher, 1920s–1935
4. The term Likelihood function for statistical model inference by Fisher, 1922
5. The term A binomial distribution with the complementary log-log link by Fisher, 1922
6. The term Exponential family (a class of distributions) with sufficient statistics by Fisher, 1934
7. The term Probit Analysis (a binomial distribution with the probit link) by Bliss, 1935

8. The term Logit for proportions by Berkson, 1944; Dyke and Patterson, 1952
9. The term Item analysis (a Bernoulli distribution with the logit link) by Rasch, 1960
10. The term Log-linear models for counts (a Poisson distribution with the log link) by Birch, 1963
11. The term an exponential distribution with the reciprocal or the log link for survival data by Feigl and Zelen, 1965; Zippin and Armitage, 1966; Glasser, 1967
12. The term Inverse polynomial (a gamma distribution with the reciprocal link) by Nelder, 1966
13. The term a synthesis of various models under the exponential family by Lindsey, 1971
14. The term generalized linear models, GLMs by Nelder and Wedderburn, 1972
15. The term Generalized Estimating Equations, GEEs, by Liang and Zeger, 1986
16. The term GAMs, Generalized additive models, by Hastie and Tibshirani, 1986, 1990; Jorgensen, 1997
17. The term Generalized Linear Mixed Models, GLMMs, by McCulloch and Searle, 2001

Thus, methods for analyzing categorical data consist of using both descriptive and inferential statistical techniques, together with statistical modeling, which has been developed by extending the principles of traditional linear models under the assumption of a normally distributed response variable to distributions within the exponential family, such as the generalized linear models, or GLMs, mentioned above.
At present, categorical data analysis can employ distributions outside the exponential family (Non-exponential Family) for model construction and for analyzing additive or nonlinear models (GAMs: Generalized additive models). There also exist generalized linear mixed models (GLMMs), which are used for various types of data: continuous, discrete, and categorical data with repeated measurements (repeated categorical response data) as well as models that consider

random effects (random-effects modeling) for response variables whose distributions are within the exponential family and outside the exponential family. The objective is to make inferences about the set of parameters that describe the structural relationships between the response variable and the various explanatory variables in a broad and comprehensive manner.

The development of models far beyond the original forms can be seen in generalized linear models (GLMs), such as logit models, probit models, log-linear models, multinomial logit models, as well as the application of GLMs to data with repeated measurements in the same sample units or experimental units, such as data collected at different time points or occasions. This also includes the use of nonlinear models mentioned above, such as generalized additive models (GAMs) and generalized linear mixed models (GLMMs), which incorporate both the fixed effects parameter vector (β) and the random effects parameter vector (u) with parameter estimation based on generalized estimating equations (GEEs). These models are used for data obtained from both random sample surveys and randomized experiments, or from both sources.

The difference between GLMs and traditional linear models is that, in the latter, the response variable is continuous and normally distributed, whereas in generalized linear models and nonlinear models, the response variable follows a distribution within the exponential family or outside the exponential family. The explanatory variables may be either continuous or categorical variables, respectively. If all variables of interest are categorical, the data will consist of a set of observed counts. The data analysis is therefore in the form of contingency table analysis, and can also be analyzed using categorical data analysis models.

Categorical data analysis may begin with preliminary methods such as examining the proportions of interest, percentages by rows or by columns of the categorical variables that correspond to the distributional characteristics or the sampling design, differences between the proportions of interest, various preliminary measures of association, the assessment of risk, the calculation of odds and odds ratios, as well as methods for constructing various statistical models. The appropriate model is selected to correspond as closely as possible to the data and theoretical conditions, and the resulting model is then used to summarize the findings.

The construction of statistical models is generally based on the distributional characteristics of the response variable or depends on the sampling schemes that describe the distribution of the observed counts obtained from the sample. The commonly used schemes are:

(1) counts or observations that follow a **Poisson distribution and are independent of one another**

(2) counts or observations that follow a **multinomial distribution and are independent of one another**

including the Bernoulli distribution and the binomial distribution as well.

In addition, other distributional patterns that are commonly encountered in experiments include the hypergeometric distribution and the negative binomial distribution, as well as other types of distributions that may fall within or outside the exponential family.

Once the distributional form is known, the model is then constructed under the distribution of the response variable and the explanatory variables, and the parameters are estimated from the probability density function of the response variable, which is a function of the response variable given the parameters under a particular distribution, such as $f(x|\theta)$ where $\theta=(\theta_1,\dots,\theta_k)$ represents the parameter vector in the model, and k represents a constant.

The estimation of parameters in categorical data analysis commonly uses the method of maximum likelihood to estimate the parameters of the model. The estimates obtained are called maximum likelihood estimates, which rely on the joint probability density function or the likelihood function. These estimators possess several properties, most of which are related to asymptotic theory and the asymptotic properties of the maximum likelihood estimator.

In addition to estimating the parameters of the model that describe the structural relationship between the variables, it is necessary to first evaluate whether the model is appropriate for the data, for example, by testing hypotheses regarding the parameters, assessing goodness of fit, analyzing residuals, and so forth. If there are multiple models, the different models should be compared in order to select the most appropriate one. Then, the results and interpretations of the chosen model should be summarized, concerning the influence of the explanatory variables on the response variable, the risk, as well as interaction effects and conditional effects of the variables based on the parameter estimates of the model.

The continuous development in both theory and applications of categorical data analysis has also included the development of statistical software packages such as SAS, S-Plus, R, Minitab, GLIM, SPSS.

1.5 **Statistical analysis techniques for response variables and explanatory variables**

As mentioned in Sections 1.2 and 1.4, categorical variables have several types and can be measured on various scales. The data analysis depends on the characteristics of the response variable and the explanatory variables. Some important analysis techniques commonly used for response and explanatory variables, according to the summary of Dobson (1990), are summarized in Table 1.9 below.

Table 1.9 Commonly used statistical analysis techniques classified according to the measurement scales of the response variable and the explanatory variable

Explanatory variable (X)	**Response variable** (Y)		
	2 categories or levels (Dichotomous or binary, Nominal)	>2 categories or levels (Polychotomous, Nominal – Interval)	Continuous
2 groups	comparison of proportions, contingency table analysis Logistic models, Loglinear models	contingency table analysis Loglinear models, Cumulative logit models, Nonparametric tests	t-tests correlation coefficient t-tests correlation coefficient Nonparametric tests
>2 groups (Nominal)	contingency table analysis Generalized logistic models, Loglinear models	Loglinear models, Polychotomous logits, Cumulative logit models, Models for matched pairs, GLMs, GAMs, GLMMs, and other mixed models	ANOVA models Analysis of variance Nonparametric tests
Continuous	Logistic models	Polychotomous logits, Cumulative logit models	Regression analysis
Mixed (Categorical & Continuous)	Logistic models	Polychotomous logits, Cumulative logit models,	ANCOVA models Analysis of covariance

	Response variable (Y)		
Explanatory variable (X)	2 categories or levels (Dichotomous or binary, Nominal)	>2 categories or levels (Polychotomous, Nominal – Interval)	Continuous
		GLMs, GAMs, GLMMs, and other mixed models	

GLMs: Generalized Linear Models, GAMs: Generalized Additive Models, GLMMs: Generalized Linear Mixed Models.

Note: The number of response variables and the number of explanatory variables mentioned above may exceed one. The analysis techniques may therefore be adjusted accordingly, and other analysis techniques may also be applicable.

However, the generalized linear model technique can be extended to multivariate generalized linear models, such as various types of complex loglinear models.

1.6 Statistical model building

Categorical data analysis includes many analytical techniques, as illustrated in Table 1.9. In addition to comparing proportions, analyzing contingency tables, and testing for independence, one may also be interested in examining the association among groups of data in contingency tables using other methods, such as parameter inference for models in order to describe the relationships among variables. This is done by estimating the parameters of **statistical models or probabilistic models**, which are models that express the relationships of each explanatory variable, or between explanatory variables, on the response variable, based on statistical theory under a specific distribution and sampling scheme, such as the normal, Bernoulli, binomial, Poisson, or multinomial distributions of the response variable, and models such as regression models, analysis of variance models, logistic regression models, simple (two-dimensional) and complex (multidimensional) log-linear models, logit models, cumulative logit models, and so on.

The principles of constructing statistical models consist of processes that can be summarized in 6 steps as follows:

1. Specify the probability distribution of the response variable
2. Specify the explanatory variables that are relevant and potentially relevant according to expert knowledge
3. Construct reasonable (plausible) models or equations that can describe the essential characteristics of the response variable and the explanatory variables
4. Estimate the various parameters of the model in Step 2
5. Select the most appropriate model
6. Assess the model, such as its goodness of fit, residuals, and hypothesis testing
7. Interpret the model, evaluate the results, and summarize the findings

Readers should also review certain basic statistical knowledge, such as the probability distributions of random variables and of statistics, hypothesis testing, introductory contingency table analysis, Chi-square tests $(\chi^2 - \text{test})$, measures of association, regression analysis, analysis of variance, and linear models. In addition, having basic mathematical knowledge such as vectors and matrices, algebra and calculus will help in understanding the data, statistics, and various results more quickly.

1.7 Probability distributions for categorical data and related sampling schemes

According to the principles of constructing statistical models in Section 1.6, the first matter that should be considered is the distributional characteristics of the response variable and the type of explanatory variables. Other aspects should then be examined in order to construct a model based on theoretical reasoning that can be explained and verified further.

In categorical data analysis using statistical models such as generalized linear models (GLMs), the probability distribution of the response variable is generally based on count data, for which the commonly used probability distributions belong to the exponential family, such as:

1. Bernoulli distribution
2. Binomial distribution
3. Beta binomial distribution
4. Poisson distribution
5. Negative binomial distribution
6. Multinomial distribution
7. Product Multinomial distribution
8. Hypergeometric distribution
9. Normal distribution
10. Gamma distribution
11. Inverse Gaussian distribution
12. Exponential distribution

For more complex models such as GLMMs (Generalized Linear Mixed Models), which are now increasingly used, and GAMs (Generalized Additive Models), which are still used only in certain fields, probability distributions outside the exponential family (non-exponential family) may be employed.

Categorical data generally arise from different sampling procedures or experiments (different sampling frameworks). Therefore, inference requires knowing the distributional characteristics of such data in order to serve as a basis for selecting appropriate methods of data analysis and for conducting correct statistical inference. The distributional characteristics of data, especially for categorical response variables, are commonly those in the exponential family, such as Bernoulli, Binomial, Multinomial, Poisson, and Hypergeometric distributions, and so on, which are widely used for analyzing categorical data using various GLMs. These will be discussed in the next chapter.

Examples of distributions commonly encountered in practice can be observed from contingency tables of size $(I \times J)$ of any kind. If the sampling or experiment does not specify the total sample size, and the sample size is random (n random), the distribution of the observed counts in the contingency table will be Poisson. But if the sample size is fixed (n fixed), the distribution will be Multinomial. If only the row or the column marginal totals (Fixed row or column marginal totals) are fixed on one side, while the other side is random, the distribution of the data will be Product multinomial, and if both the row and column marginal totals are fixed, the distribution of the data will be Hypergeometric.

At present, statisticians have developed the use of models that are more complex than GLMs, such as GLMMs and GAMs, which can employ distributions beyond those in the exponential family. Although statistical software is available to assist with computation, there remain difficulties in interpreting these models, which are considerably more complex than GLMs. Therefore, they are still used less widely than GLMs

1.7.1 Bernoulli Distribution

The Bernoulli distribution is used to describe an experiment or sampling in which the random variable has two possible outcomes: success or failure, independent of one another. Such an experiment is called a Bernoulli trial, with the probability of success equal to p, and the entire process is called a Bernoulli process.

The probability mass function of the Bernoulli distribution is

$$p(y) = p^y q^{(1-y)}; \ y = 0,1$$

or $$p(y) = \begin{cases} p, & y = 1 \\ q, & y = 0 \end{cases}$$

where $0 \le p \le 1, \ q = 1 - p$

and $E(Y) = p, \ \text{var}(Y) = pq$

1.7.2 Binomial distribution

Let $Y_1, Y_2, \ldots, Y_n$ represent Bernoulli random variables that are identically distributed and independent (i.i.d. or independent, identically distributed random variables) with probability of success $(Y = 1)$ equal to p the random variable $Y = Y_1 + Y_2 + \ldots + Y_n$ has a Binomial distribution, where p is fixed or the same for all units of the experiment, and the $\{Y_i\}$ are independent of one another.

The Binomial distribution is the distribution of a random variable that is the sum of successes in a Bernoulli process of n trials, or is the result of drawing a sample of size n from an infinite population in which each unit is selected independently and has the same probability of success p. Thus, the sum of the number of successes, $Y = \sum_{i=1}^{n} Y_i$ follows a Binomial distribution, or is denoted by $Bin(n, p)$ to represent the Binomial distribution with parameter (n, p)

The probability mass function of the Binomial distribution is

$$P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0,1,2,\ldots,n$$

Where $\binom{n}{y}$ represents the binomial coefficient, which is equal to

$$\frac{n!}{y!(n-y)!}, \; y \le n$$

From each Bernoulli trial of $Y = 0$ or 1 obtain

$$E(Y_i) = E(Y_i^2) = 1 \times p + 0 \times (1-p) = p$$

$$\therefore \quad E(Y_i) = p$$

$$\text{var}(Y_i) = p(1-p)$$

Therefore, the Binomial distribution with $Y=\sum_{i=1}^{n} Y_i$ has the following mean and variance: $\mu = E(Y) = np$ and $\text{var}(Y) = np(1-p)$ and this distribution converges to normality when … is large for a fixed p

Note: Y represents a Binomial random variable with parameters n and p

(1) If $n=1$ Y is a Bernoulli random variable with probability of success p

(2) When $n \to \infty$ If $np \geq 5$ and $np(1-p) \geq 5$, then y is approximately distributed as a normal distribution with $\mu = np$ and $\sigma^2 = np(1-p)$

(3) When $n \to \infty$ If $p < 0.1$ and $np < 10$, then y is approximately distributed as a Poisson distribution with $\lambda = np$

(4) Let $Y_1, \ldots, Y_k$ be independent Binomial random variables with parameters n_i and p , respectively. The random variable $Y = Y_1 + \ldots + Y_k$ has a Binomial distribution with parameters $n = n_1 + \ldots + n_k$ and p

If the sample is drawn using sampling without replacement from a finite population such as selecting 10 students from a class of 20 and observing data such as male or female, the data analysis can rely on the Hypergeometric distribution instead of the Binomial distribution, because the categorical data in this case are not independent of one another.

1.7.3 Multinomial distribution

In a two-way contingency table of size n with independent and identically distributed data, one may be interested in observations belonging to more than two categories.

Let $y_{ij}=1$ if the observation or experiment in the i falls into category $j,\ j=1,2,\ldots,k$ and $y_{ij}=0$ for other categories. Thus, $Y_i=(y_{i1},y_{i2},\ldots,y_{ik})$ is a Multinomial trial with specified $\sum_{j}^{k} y_{ij}=1$, for example, $k=4$ where $(y_1,y_2,y_3,y_4)=(0,1,0,0)$ represents the outcome of category 2 for Y with 4 categories.

Let $n_j=\sum_{j}^{n} y_{ij}$ represent the observed value or the number of outcomes in category j. The count $(n_1,n_2,n_3,\ldots,n_k)$ follows a Multinomial distribution.

Let $p_j=P(Y_{ij}=1)$ represent the probability of the outcome in category j.

The probability mass function of the Multinomial distribution for the counts $n_j,\ j=1,2,\ldots,k$ is

$$P(n_1,n_2,n_3,\ldots,n_k) = \left(\frac{n!}{n_1!n_2!\ldots n_k!}\right)p_1^{n_1}\ldots p_k^{n_k}$$

Where $\sum_j n_j = n,$

$$n_k = n-(n_1+\ldots+n_{k-1})$$

$$E(n_j)=np_j,$$

$$\text{Var}(n_j)=np_j(1-p_j),$$

$$\text{Cov}(n_i,n_k)=-np_jp_k,\ i\neq j=1,2,\ldots,k$$

Note:

(1) The marginal distribution of the Multinomial distribution n_i is Binomial with parameters n and p_i

(2) If $k=2$ and $p_1=p$ **, the Multinomial random variable** n_i becomes a **Binomial random variable** with parameters n and p (the

Binomial distribution is a special case of the Multinomial distribution when $k=2$)

(3) **The Multinomial distribution can be used in cases where the probabilities** $p_i,\ i=1,2,\ldots,k$ are equal or unequal, and it is applied in sampling and advanced statistical analysis.

1.7.4 Product or Independent multinomial distribution

In a two-way contingency table, if the random variable representing the response variable Y is placed in the columns, while the explanatory variable X is placed in the rows, let n_{ij} represent the count of observations in category j of at level i of X, $j=1,2,\ldots,k,$ $i=1,2,\ldots,r$.

Assume that n_i represents the observed value of y at level i of X, where n_i are independent of each other and have probability equal to $\{p_{1|i}, p_{2|i},\ldots, p_{k|i}\}$.

Thus, the counts $\{n_{ij},\ j=1,2,\ldots,k\}$ follow independent Multinomial distributions.

The probability mass function of the Product multinomial is

$$P(n_{ij}) = \left(\frac{n_i}{\prod_{j=1}^{k} n_{ij}}\right)\prod_{j=1}^{k} P_{1|i}^{n_{ij}}$$

and the above $P(n_{ij})$ is called the "Independent or Product Multinomial Probability Mass Function".

1.7.5 Poisson distribution

The Poisson distribution for the counts y_i that occur randomly within a unit of time, or a specified area or distance (a specific time interval, given region, or space), independently of one another, with a small probability (rare event).

The probability mass function (by Poisson, 1837) is

$$P(y) = \frac{e^{-\mu}\mu^{y}}{y!}, \; y = 0,1,2,\ldots$$

Where $E(Y) = \mathrm{Var}(Y) = \mu$

For a random variable that follows a Poisson distribution (in this case n is not a constant), the Poisson distribution approaches the normal distribution when μ increases, and it can also be approximated by the Binomial distribution when n is large and π is small, with mean and variance equal to $n\pi$ for example, during the Songkran festival, each person drives independently to other provinces, with the probability of having an accident equal to 0.00003. If the population is 24 million people using vehicles during the festival period, the number of accidents that occur follows a Binomial distribution or $bin(224,000,000, 0.00003)$, or is approximated by a Poisson distribution as $Poi(720)$.

Note: Let Y represent a Poisson random variable with parameter λ

(1) When $\lambda \to \infty$, Y it is approximately distributed as a normal distribution with $\mu = \lambda$ and $\sigma^2 = \lambda$

(2) Let $Y_1, \ldots, Y_n$ represent independent random variables with parameters λ, respectively. The random variable $Y = Y_1 + \ldots + Y_n$ follows a Poisson distribution with parameter $\lambda = \lambda_1 + \ldots + \lambda_n$

(3) The waiting time between occurrences of successive Poisson random variable values follows an Exponential distribution (The waiting time between Poisson arrivals is exponentially distributed)

1.7.6 Hypergeometric distribution

From the example of contingency tables under Poisson or Binomial distributions that are independent, if the observed marginal totals of the categorical variables are fixed or conditioned to be constant, then the probability of success p is no longer constant. Therefore, the number of successes (Y) has a new type of data distribution, namely the Hypergeometric distribution. For example, in a population of size N units, consisting of M units of success and $N-M$ units of failure, if a sample of size n units is drawn without replacement, the probability of success (Y) will not remain constant.

The probability mass function of the Hypergeometric distribution is

$$P(y) = \frac{\binom{M}{y}\binom{N-M}{n-y}}{\binom{N}{n}}, \quad y = 0,1,\ldots,n$$

Where $y \le M,\ n-y \le N-M,\ 1 \le n \le N,\ 1 \le M \le N,\ N > 0$ is a positive integer

and $E(Y) = \frac{nM}{N}$

$$\mathrm{Var}(Y) = \left(\frac{N-n}{N-1}\right)\frac{nM}{N}\left(1-\frac{M}{N}\right)$$

Note: Let Y represent a Hypergeometric random variable with parameters $n,\ M$ and N

(1) When $N \to \infty$ if $\frac{n}{N} < 0.1$

Then Y has a Binomial distribution with parameters n and $p = \frac{M}{N}$

(2) When $n,\ M$ and $N \to \infty$ if $\frac{M}{N}$ is small,

Then Y is approximately distributed as a Poisson distribution with parameter $\lambda = \frac{nM}{N}$

(3) For example, in a contingency table of size (2×2) with n_1 and n_2 fixed the distribution of $n_{ij},\ j=1,2$ depends on n_{11}

The probability mass function of the Hypergeometric distribution in a contingency table of size (2×2) is

$$P(n_{ij}) \ = \ \frac{\binom{n_1}{n_{11}}\binom{n_2}{n_1-n_{11}}}{\binom{n}{n_1}} \ = \ \frac{n_1!n_2!n_1!n_2!}{n!n_{11}!n_{12}!n_{21}!n_{22}!}$$

1.7.7 Negative binomial distribution

From the negative binomial distribution with probability of success equal to p , if one is interested in the number of failures before observing the k then the distribution of the number of failures (y) is a Negative binomial distribution.

The probability mass function of the negative binomial distribution is

$$P(y) \ = \ \binom{y+k-1}{k-1} p^k (1-p)^y$$

Where $y = 0,1,\dots\ ,\ k=1,2,\dots\ ,\ 0 \le p \le 1$

$$E(Y) = \frac{k(1-p)}{p},$$

$$\text{Var}(Y) = \frac{k(1-p)}{p^2}$$

Note: Let Y have a negative binomial distribution with parameters k and p

(1) If $k=1$ then Y follows a Geometric distribution with probability of success p

(2) When $k \to \infty$ and $p \to 1$, with $k(1-p)$ fixed, then is approximately Poisson, with mean and variance equal to $\lambda = k(1-p)$

(3) Let $y_1, y_2, \ldots, y_m$ be independent Negative binomial random variables with parameters k_i and p respectively; then the random variable has a negative binomial distribution with parameters $n = k_1 + k_2 + \ldots + k_m$ and p.

Exercises for Chapter 1 and Answers

1.1 Analyze the following data using an appropriate analytical technique by employing GLIM and SPSS. Assume that the sample data for the random response variable Y and the explanatory variables X_1, X_2, and X_3 each contain $n=10$ observations, and the data are as follows:

Y	13.8	16.9	19.6	23.0	25.8	28.9	30.8	33.7	35.8	42.0
X_1	23.5	23.6	23.9	24.6	25.2	25.8	26.3	26.5	26.9	26.9
X_2	23.8	23.3	24.9	28.3	31.4	34.1	37.0	38.3	42.3	45.0
X_3	16.1	16.9	17.8	18.9	20.0	21.4	22.8	24.2	25.3	27.1

Answer: Since all variables contain continuous data, a multiple regression model should be constructed.

<u>The GLIM commands</u> are:

$UNITS 10$DATA Y $READ 13.8 16.9 ……………. 42.0$
$DATA X1 $READ 23.5 23.6 ……………. 26.9$
$DATA X2 $READ 23.8 23.3 ……………. 45.0$
$DATA X3 $READ 16.1 16.9 ……………. 27.1$
$YVAR Y$FIT :X1 :X2 :X3 :X1+X2 :X1+X3 :X1+X2+X3$DISP E$

<u>The SPSS commands</u> are: When the data Y, $X1-X3$ are in the data file (Worksheet column 1–4), **Select: Analyze, Regression, Linear,**

enter the dependent variable Y and the independent variables $X1-X3$, then click OK.

The results are:

Table 1.1 Deviances according to models and degrees of freedom

(d.f.)

Model	**Deviances**	**d.f.**
$GM(\beta_0)$	713.8	9
$\beta_0+\beta_1X_1$	40.45	8
$\beta_0+\beta_1X_2$	19.77	8
$\beta_0+\beta_1X_3$	9.078	8
$\beta_0+\beta_1X_1+\beta_2X_2$	19.67	7
$\beta_0+\beta_1X_1+\beta_2X_3$	9.031	7
$\beta_0+\beta_1X_2+\beta_2X_3$	8.949	7
$\beta_0+\beta_1X_1+\beta_2X_2+\beta_3X_3$	8.712	6

From Table 1.1, Regression ANOVA Table 1.2 can be constructed as follows (SPSS will also provide Table 1.2):

Table 1.2 Regression ANOVA

SOV	**d.f.**	**SS**	**MS**
Regression	3	705.1	253.3
Residual	6	8.7	1.45
Total	9	713.8	

From Table 1.2, the calculation of $F_{ratio}=\frac{235.03}{1.45}>F_{3,6,.05}$ shows that it is statistically significant at the 0.05 level. Therefore, the regression model can explain the dependent variable with statistical significance at the 0.05 level. The parameter estimates in the regression model are:

	estimate	**s.e.**	**Parameter**
1.	-37.25	32.55	% GM
2.	0.6516	1.613	X_1
3.	-0.2604	0.5552	X_2
4.	2.676	0.974	X_3

Thus, the multiple regression model has the following form:

$$\hat{Y} = -37.25+0.6516X_1-0.2604X_2+2.676X_3$$

which may be used to predict values of the dependent variable at any specified values of the variables X ✈

1.2 The model in Question 1.1 can be selected using the Stepwise method with SPSS. That is, after selecting Regression > Linear, enter the dependent variable and the independent variables $X1-X3$, then in the same window choose Stepwise instead of Enter, and select OK. The regression model with parameter estimates will be produced by SPSS as required. ✈

1.3 The model in Question 1.1 can be selected by comparing the differences, or ΔDev. (Change of Deviances), between pairs of models using GLIM, and comparing ΔDev with $\chi^2_{\Delta df,\alpha} = \chi^2_{\Delta df,0.05}$ If rejected, the shorter model is rejected, and the longer model is accepted accordingly. In addition, the Deviances in Table 1.1 can be used to construct the ANOVA table as shown in Table 1.2. ✈

1.4 From the following data table, construct an ANOVA table using GLIM and SPSS for testing whether there are differences among the 4 types of treatments at the 0.05 level of statistical significance.

T1	**T2**	**T3**	**T4**
54.0	51.0	53.1	50.7
50.8	52.2	42.0	53.8
55.0	50.0	41.5	54.9
47.1	53.9	50.8	55.8
47.6	57.5	50.0	51.2
50.0	55.2	45.8	56.7

Answer: The model of interest is the One-way ANOVA.
<u>The GLIM commands</u> are:
$UNIT 24 $DATA Y$
$READ 54.0 50.8 55.0 47.1 47.6 50.0 51.0 … 56.7$

$DALC T = %GL(4,6)$
$FACT T 4$YVAR Y$FIT : T $DISP E$

<u>The SPSS commands</u> are:
Select Analyze, Compare Means, One-Way ANOVA, enter the dependent variable as and the factor as X, where Y represents the data stored in C1, and X represents the numeric codes indicating the treatment groups 1, 2, 3, and 4 for the values of Y, stored in C2 , then click OK.

The result is:

cycle	deviance	d.f.
1	404.2	23
cycle	deviance	d.f.
1	238.5	20

The ANOVA table is Table 1.3

Table 1.3 One-way ANOVA

SOV	**d.f.**	**SS**
Treatment	3	165.7
Error	20	238.5
Total	23	404.2

From Table 1.3, the calculation of $F_{ratio} = \frac{165.7}{238.5} < F_{3,6,.05}$ shows that it is not statistically significant at the 0.05 level. Therefore, there is not enough information to conclude that the 4 treatments differ from each other statistically at the 0.05 level of significance. ✈

1.4 Analyze the differences among treatments from the ANOVA table in Example 1.4, and determine whether it is necessary to further examine pairwise treatment differences. Demonstrate the procedure using SPSS.

Answer <u>The SPSS commands</u> are:

Select Analyze > Compare Means > One-way ANOVA
Enter the dependent variable as Y and the factor as X , then select OK.

If further comparisons are required, select Post Hoc and choose one of the methods for pairwise treatment comparisons, such as LSD or Tukey, then select OK. The output will display all possible pairwise comparisons of treatment means.

However, in the overall analysis, from the results in Question 1.4, it was found that there is not enough information to conclude that the four treatments differ statistically at the 0.05 level.
Therefore, it is not necessary to perform additional pairwise comparisons of treatments, because all pairs will yield statistically non-significant differences. ✈

1.5 From measurements obtained to study experimental data for testing 3 types of materials under 4 levels of humidity, suppose the experimental results yield the data shown in the following table:

		Humidity (H)			
		1	2	3	4
Material (M)	1	39.00	33.10	33.80	33.00
		42.80	37.80	30.70	32.90
	2	36.90	27.20	29.70	28.50
		41.00	26.80	29.10	27.90
	3	27.40	29.20	26.70	30.90
		30.30	29.90	32.00	31.50

Analyze the data using GLIM to assist with computation.

Answer: The model of interest is the Two-way ANOVA with interaction model.

<u>The GLIM commands</u> are:

$UNIT 24$READ 39.0 42.8 36.9 41.0 27.4 30.30 33.10 37.80 … 31.50$

$CALC M = %GL(3,2) : H = %GL(4,6)$

$FACT M 3 H 4$YVAR Y$FIT : M : +H : +H.P$

<u>The SPSS commands</u> are as follows:

Select Analyze > General Linear Model > Univariate, and enter the dependent variable Y which is stored entirely in C1, Enter the factors under Fixed Factors as H and M, representing the levels of humidity and the types of materials stored in C2 and C3, respectively, then select OK.

The results are:

cycle	deviance	d.f.
1	470.9	23
cycle	deviance	d.f.
1	328.9	21
cycle	deviance	d.f.
1	184	18
cycle	deviance	d.f.
1	50.59	12

From the above results, the ANOVA table can be constructed as shown in Table 1.4.

Table 1.4 Two-way ANOVA with Interaction

SOV	**d.f.**	**SS**	**F-ratio**
Material	2	142.6	16.91
Humidity	3	143.8	11.37
Interaction effect	6	133.9	5.29
Error	12	50.6	
Total	23	470.9	

Since $F_{ratio} = 2.64 < F_{6,12,.05} = 2.99$

Therefore, the interaction effect is statistically significant at the 0.05 level (F = 5.29 > F-critical(0.05, 6, 12) = 3.00).

Since the interaction effect is significant, the main effects of material and humidity should be interpreted with caution. Both factors also show significant F-ratios: F(Material) = 16.91 > F-critical(0.05, 2, 12) = 3.89, and F(Humidity) = 11.37 > F-critical(0.05, 3, 12) = 3.49.

It is found that

$F_{ratio} = 2.82 < F_{2,12,.05} = 3.88$. Therefore, the 3 types of materials differ significantly at the 0.05 level of statistical significance.

$F_{ratio} = 2.84 < F_{3,12,.05} = 3.49$ Therefore, the humidity levels differ significantly at the 0.05 level of statistical significance.